

CLEARFIELD-LAWRENCE TOWNSHIP, PA, USA

WMO: 725119

Lat: 41.047N

Lon: 78.412W

Elev: 1516

StdP: 13.91

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54792

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	6.5	-7.4	3.9	5.1	-2.9	5.0	8.5	23.0	25.6	20.5	26.1	6.9	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	87.4	70.7	84.7	69.3	82.3	68.2	73.6	82.1	72.2	79.9	70.8	78.1	8.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	122.2	77.1	70.0	116.8	75.7	68.5	110.7	74.2	38.2	82.2	36.7	79.7	35.6	78.1	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.3	15.4	-6.1	91.9	6.2	3.7	-10.6	94.5	-14.3	96.7	-17.8	98.8	-22.3	101.4		
(5)			-6.7	76.3	6.1	1.6	-11.1	77.5	-14.7	78.5	-18.2	79.4	-22.6	80.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	24.6	26.8	36.3	48.3	58.8	65.9	70.4	68.8	62.3	50.9	39.9	30.6		
	DBStd	18.04	11.04	9.87	10.32	9.27	8.26	5.82	5.05	4.99	6.74	8.56	8.57	8.87		
	HDD50	3060	789	651	442	139	18	0	0	0	2	95	321	603		
	HDD65	6430	1253	1070	889	503	226	61	11	21	130	446	754	1066		
	CDD50	2601	1	1	18	90	291	476	632	582	370	122	16	2		
	CDD65	496	0	0	0	3	33	87	178	139	48	8	0	0		
	CDH74	4197	0	0	3	85	379	717	1458	1068	431	55	1	0		
	CDH80	1087	0	0	0	12	75	165	436	280	110	9	0	0		
	WSAvg	6.3	7.6	7.6	7.4	7.4	6.0	5.4	4.8	4.5	4.8	6.0	6.5	7.2		
	PrecAvg	42.0	2.8	2.7	3.5	3.4	4.3	4.4	4.1	3.5	3.7	3.0	3.5	3.1		
	PrecMax	50.9	6.0	5.6	6.5	6.3	8.6	10.9	10.2	6.7	8.7	6.9	9.2	5.0		
(15) Precipitation	PrecMin	30.6	0.8	0.6	0.9	0.7	0.8	0.4	1.5	0.8	0.9	0.4	0.9	1.3		
	PrecStd	5.0	1.5	1.3	1.6	1.5	1.8	2.4	1.7	1.5	1.7	1.5	1.7	1.2		
	0.4%	DB	57.5	58.6	71.9	81.7	85.8	87.9	91.6	89.7	87.9	80.0	68.3	59.3		
	2%	MCWB	51.0	50.0	56.6	62.6	67.8	70.6	72.3	71.6	70.3	67.2	56.7	54.9		
		DB	51.2	50.5	64.8	76.0	82.3	84.5	87.7	86.0	83.3	74.0	62.7	53.3		
(21) Mean Coincident Wet Bulb Temperatures	5%	MCWB	46.9	43.6	52.3	57.8	65.1	69.3	70.9	70.4	68.1	63.0	53.6	49.3		
		DB	44.8	45.4	58.6	70.6	78.9	81.5	84.5	83.1	79.2	70.0	58.5	47.5		
	10%	MCWB	41.5	40.0	48.9	54.7	63.4	68.0	69.8	69.3	66.6	61.2	51.1	43.5		
		DB	39.7	40.5	53.3	65.3	74.4	78.3	81.9	80.4	74.9	65.2	54.3	43.1		
	10%	MCWB	36.6	35.9	46.5	52.1	61.6	66.0	68.9	67.8	64.4	58.5	48.4	39.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	52.5	58.9	64.5	70.4	73.2	76.2	75.5	74.4	69.6	60.1	55.6		
		MCDB	57.4	55.9	68.3	76.0	81.4	82.3	85.6	84.0	82.8	76.9	64.8	58.2		
	2%	WB	47.0	45.5	55.3	60.7	68.2	71.5	74.0	73.2	71.0	66.4	55.9	50.1		
		MCDB	49.5	49.8	61.8	70.3	77.6	80.4	82.9	80.9	78.0	71.4	60.5	52.3		
	5%	WB	41.8	40.5	51.3	58.0	66.3	70.1	72.4	71.5	68.9	62.9	52.5	44.4		
(31) Mean Daily Temperature Range	5%	MCDB	45.0	44.6	57.0	66.6	74.8	78.3	80.4	78.0	75.4	67.4	57.0	46.9		
		WB	36.9	36.2	46.2	54.7	64.1	68.5	70.9	70.1	66.9	59.3	49.0	40.2		
	10%	MCDB	39.3	40.3	52.1	62.4	71.4	75.3	78.1	76.6	72.4	65.0	53.6	42.8		
		WB	36.9	36.2	46.2	54.7	64.1	68.5	70.9	70.1	66.9	59.3	49.0	40.2		
	10%	MCWB	39.3	40.3	52.1	62.4	71.4	75.3	78.1	76.6	72.4	65.0	53.6	42.8		
(35) Clear-Sky Solar Irradiance	taub	MDBR	14.4	15.8	18.5	22.1	21.9	20.6	21.0	20.9	21.0	19.0	16.9	12.7		
		MCDBR	20.0	22.6	27.8	31.5	27.2	24.4	24.0	24.7	26.1	25.8	24.8	19.2		
	Ebn at Noon	MCWBR	16.7	17.1	17.1	16.3	12.8	11.6	10.1	10.7	12.4	14.5	16.5	15.9		
		MCDBR	19.2	20.4	24.5	25.8	22.1	20.5	20.3	19.7	20.6	20.8	20.7	17.8		
	Edh at Noon	MCWBR	17.0	16.9	17.2	15.4	11.1	10.7	9.6	9.5	10.5	12.8	15.9	16.0		
(40) All-Sky Solar Radiation	RadAvg	taub	0.310	0.329	0.351	0.374	0.409	0.457	0.462	0.455	0.375	0.366	0.323	0.301		
		taud	2.387	2.308	2.350	2.336	2.257	2.145	2.142	2.153	2.363	2.381	2.480	2.467		
	Ebn at Noon	taub	267	278	285	285	276	262	259	257	273	261	261	262		
		taud	26	33	36	38	43	48	48	46	35	30	24	23		
	RadStd	RadAvg	487	746	1059	1417	1610	1763	1793	1603	1302	867	581	391		
(45)		RadStd	38	89	103	142	162	149	129	95	155	95	74	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	N/A

Nomenclature: See separate page